Sentiment Analysis Using Traditional Machine Learning

Objective

Practice Basic NLP tasks

Practical Information

Topic: Text Preprocessing, TF-IDF and Machine Learning

Duration: 1 Hour

Programming Language: Python

Libraries: Pandas, NumPy, NLTK, Scikit-learn, Matplotlib, Seaborn

Dataset : https://archive.ics.uci.edu/dataset/331/sentiment%2Blabelled%2Bsentences

Tasks

1. Download and load the dataset.
2. Perform basic text preprocessing
    - Convert text to lowercase.
    - Remove unnecessary characters and punctuation.
    - Remove extra whitespace.
    - Create a clean-text column.
    - Compare original and preprocessed text.
3. Explore the text data
    - Display positive and negative examples.
    - Identify frequently occurring words.
    - Observe differences between positive and negative reviews.
4. Split the dataset
5. Create BOW representation
6. Train a ML model (SVM, Logistic Regression etc...)
7. Evaluate the model's performance. (Accuracy, Precision, Recall, F1)
8. Test the model with new sentence.